

COMPANION FOR CHAPTER 12

OF

**FUNDAMENTALS
OF PROBABILITY**

WITH STOCHASTIC PROCESSES

FOURTH EDITION

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Chapter 12

Stochastic Processes

12A ADDITIONAL EXAMPLES

Example 12a Buses of tourists arrive at a museum at a Poisson rate of λ per hour. For $15 \leq i \leq 34$, each bus has i passengers with probability p_i , $\sum_{i=15}^{34} p_i = 1$. The time each tourist spends at the museum is a random period with distribution function G independent of all other tourists. Find the expected value of the total number of tourists who have visited and left the museum by time t .

Solution: Let $X(t)$ be the total number of tourists who have visited and left the museum by time t . Let $X_i(t)$ be the total number of tourists from the i th bus who have visited and left the museum by time t . Let S_{ij} be the time that the j th tourist on the i th bus spends in the museum, and define $Y_{ij} = 1$, if the j th tourist on the i th bus has visited and left the museum by time t ; $Y_{ij} = 0$, otherwise. Conditioning on $N(t)$ yields

$$\begin{aligned} E[X(t)] &= \sum_{n=0}^{\infty} E[X(t) \mid N(t) = n] P(N(t) = n) \\ &= \sum_{n=0}^{\infty} E\left[\sum_{i=1}^n X_i(t) \mid N(t) = n\right] P(N(t) = n) \\ &= \sum_{n=0}^{\infty} \left[\sum_{i=1}^n E[X_i(t)]\right] P(N(t) = n). \end{aligned}$$

Let N_i be the number of tourists on the bus i . Let T_i be the arrival time of the i th bus. Conditioning on T_i , which is a uniform random variable over the interval $(0, t)$, yields

$$E[X_i(t)] = \int_0^t E[X_i(t) \mid T_i = x] \frac{1}{t} dx.$$

But, by Wald's equation,

$$\begin{aligned}
 E[X_i(t) \mid T_i = x] &= E\left[\sum_{j=1}^{N_i} Y_{ij}(t) \mid T_i = x\right] \\
 &= E(N_i)E[Y_{ij}(t) \mid T_i = x] \\
 &= \left(\sum_{j=15}^{34} jp_j\right)P(S_{ij} < t - x) \\
 &= \left(\sum_{j=15}^{34} jp_j\right)G(t - x).
 \end{aligned}$$

Therefore, letting $u = t - x$, we have

$$\begin{aligned}
 E[X_i(t)] &= \int_0^t \left(\sum_{j=15}^{34} jp_j\right)G(t - x)\frac{1}{t}dx \\
 &= \left(\sum_{j=15}^{34} jp_j\right)\frac{1}{t} \int_0^t G(u) du.
 \end{aligned}$$

Putting all these together, we obtain

$$\begin{aligned}
 E[X(t)] &= \sum_{n=0}^{\infty} \left[n \left(\sum_{j=15}^{34} jp_j\right) \frac{1}{t} \int_0^t G(u) du \right] P(N(t) = n) \\
 &= \left[\left(\sum_{j=15}^{34} jp_j\right) \frac{1}{t} \int_0^t G(u) du \right] \cdot \left[\sum_{n=0}^{\infty} n P(N(t) = n) \right] \\
 &= \left(\sum_{j=15}^{34} jp_j\right) \frac{1}{t} \int_0^t G(u) du \cdot E(N(t)) \\
 &= \left(\sum_{j=15}^{34} jp_j\right) \frac{1}{t} \int_0^t G(u) du \cdot \lambda t \\
 &= \lambda \left(\sum_{j=15}^{34} jp_j\right) \int_0^t G(u) du. \quad \blacklozenge
 \end{aligned}$$

Example 12b (An Inventory System with Back Orders Allowed) Suppose that for a certain product sold by a company, customer demands occur at a Poisson rate of λ per unit of time. Suppose that in the inventory of the product, whether there is shortage or there are

no shortages, for each customer demand, a replenishment order of the same size is placed so that the inventory level remains the same. When customer demands occur, if there is no shortage, the demand is met immediately. However, if there is shortage, the demand will be back-ordered and will be met when the replenishment order arrives. Such inventory models in which back orders are allowed and all customers accept late delivery can be modeled as an $M/G/\infty$ queueing system. In this model, “customers” are the replenishment orders and the service time of each “customer”, a replenishment order, is the **lead time** of the order. That is, the time interval which begins when the replenishment order is placed and ends when the replenishment order will arrive. We assume that the lead times are independent variables identically distributed each with distribution function G . Furthermore, we assume that the lead times are independent of the times customer demands occur. ♦

Example 12c Recall that an $M/G/\infty$ queueing system is an infinite-server system in which customers arrive according to a Poisson process with rate λ , service times are independent random variables with distribution function G , and service times and the arrival process are independent. Suppose that at time 0 there are no customers in an $M/G/\infty$ queueing system. Find the probability mass function of the number of customers present in the system at time t .

Solution: Let $X(t)$ be the number of customers in the system at time 0. Let $N(t)$ be the total number of customers who have arrived at the system at or prior to t . Note that $N(t)$ is the sum of $X(t)$ and the number of customers in $[0, t]$ who have already left the system. We are given that $X(0) = 0$. Conditioning on $N(t)$, we have

$$\begin{aligned} P(X(t) = i) &= \sum_{n=i}^{\infty} P(X(t) = i \mid N(t) = n) P(N(t) = n) \\ &= \sum_{n=i}^{\infty} P(X(t) = i \mid N(t) = n) \frac{e^{-\lambda t} (\lambda t)^n}{n!}. \end{aligned}$$

Given that $N(t) = n$, the arrival times of the n customers who arrived in $[0, t]$, $T_1, T_2, \dots, T_n$, are independent uniform random variables over $[0, t]$. For $1 \leq k \leq n$, let S_k be the service time of the k th customer entered the system. Let A_k be the event that the k th customer is still in the system at t . By conditioning on T_k ,

$$\begin{aligned} P(A_k) &= \int_0^t P(A_k \mid T_k = x) \frac{1}{t} dx = \int_0^t P(S_k > t - x) \frac{1}{t} dx \\ &= \frac{1}{t} \int_0^t [1 - G(t - x)] dx = \frac{1}{t} \int_0^t [1 - G(u)] du, \end{aligned}$$

where the last equality follows by making the substitution $u = t - x$. As expected, $P(A_k)$ does not depend on k . Let $p(t)$ be the probability that any one of the customers who entered the system in $[0, t]$ is still in the system at t . We have

$$p(t) = \frac{1}{t} \int_0^t [1 - G(u)] du,$$

which yields

$$P(X(t) = i \mid N(t) = n) = \binom{n}{i} [p(t)]^i [1 - p(t)]^{n-i}.$$

Therefore,

$$\begin{aligned} P(X(t) = i) &= \sum_{n=i}^{\infty} \binom{n}{i} [p(t)]^i [1 - p(t)]^{n-i} \frac{e^{-\lambda t} (\lambda t)^n}{n!} \\ &= \frac{[\lambda t p(t)]^i}{i!} e^{-\lambda t} \sum_{n=i}^{\infty} \frac{[\lambda t (1 - p(t))]^{n-i}}{(n-i)!} \\ &= \frac{[\lambda t p(t)]^i}{i!} e^{-\lambda t} \sum_{j=0}^{\infty} \frac{[\lambda t (1 - p(t))]^j}{j!} \\ &= \frac{[\lambda t p(t)]^i}{i!} e^{-\lambda t} \cdot e^{\lambda t [1 - p(t)]} \\ &= \frac{e^{-\lambda t p(t)} [\lambda t p(t)]^i}{i!}. \end{aligned}$$

So $X(t)$ is a Poisson random variable with parameter $\lambda p(t) = \frac{\lambda}{t} \int_0^t [1 - G(u)] du$. As an application, in the inventory model of Example 12b, this result implies that the number of outstanding replenishment orders at time t is Poisson with rate $\frac{\lambda}{t} \int_0^t [1 - G(u)] du$. ♦

Example 12d At a higher education institution, students are caught plagiarizing at a Poisson rate of λ per month. For each case of plagiarism, a hearing process begins when the student is caught and ends when a hearing is convened in the Dean's office and an appropriate punishment is determined. Suppose that the duration of hearing processes are identically distributed and independent with distribution function G . Furthermore, suppose that the duration of hearing processes and the Poisson process according which the students are caught plagiarizing are independent. At this institution, what is the probability that, at time t , there are exactly n unsettled plagiarism cases?

Solution: This problem can be modeled as an $M/G/\infty$ queueing system in which arrivals are plagiarizing cases caught and service times are the durations of hearing processes. Let $X(t)$ be the number of unsettled plagiarism cases by time t . Then, by Example 12c, $X(t)$ is a Poisson process with parameter

$$\mu = \frac{\lambda}{t} \int_0^t [1 - G(u)] du.$$

Therefore,

$$P(X(t) = n) = \frac{e^{-\mu t} (\mu t)^n}{n!}. \quad \blacklozenge$$

Example 12e Passengers arrive at a certain train station according to a Poisson process with rate λ and wait for a train to arrive. Trains arrive at the same station, beginning at time 0, every T minutes. Each time a train arrives, all the passengers waiting at the station will board the train. The train then immediately leaves the station. If there are no passengers waiting at the station, the train will not wait until passengers arrive. For $i \geq 1$, find the expected value of the sum of the waiting times of all of the customers who will take the train arriving at time iT .

Solution: Let W be the sum of the waiting times of all of the customers who will take the train arriving at time T . It should be clear that, for $i \geq 1$, $E(W)$ is the desired quantity. For $t \geq 0$, let $N(t)$ be the number of customers who have arrived at the station at or prior to t . Given $N(T) = n$, for $1 \leq i \leq n$, let S_i be the time of the i th arrival. We have

$$\begin{aligned} E(W) &= \sum_{n=0}^{\infty} E(W \mid N(T) = n) P(N(T) = n) \\ &= \sum_{n=1}^{\infty} E(W \mid N(T) = n) P(N(T) = n), \end{aligned}$$

where

$$\begin{aligned} E(W \mid N(T) = n) &= E[(T - S_1) + (T - S_2) + \cdots + (T - S_n)] \\ &= E[nT - (S_1 + S_2 + \cdots + S_n)] \\ &= nT - E(S_1 + S_2 + \cdots + S_n). \end{aligned}$$

Now let $X_1, X_2, \dots, X_n$ be n randomly and independently selected points from the interval $[0, T]$. Let $X_{(1)}, X_{(2)}, \dots, X_{(n)}$ be the order statistics of $X_1, X_2, \dots, X_n$. Then

$$\begin{aligned} E(S_1 + S_2 + \cdots + S_n) &= E[X_{(1)} + X_{(2)} + \cdots + X_{(n)}] \\ &= E(X_1 + X_2 + \cdots + X_n) \\ &= nE(X_1) = n\frac{T}{2}. \end{aligned}$$

Hence

$$\begin{aligned} E(W) &= \sum_{n=1}^{\infty} E(W \mid N(T) = n) P(N(T) = n) \\ &= \sum_{n=1}^{\infty} \left(nT - \frac{nT}{2} \right) \frac{e^{-\lambda T} (\lambda T)^n}{n!} \\ &= \frac{T}{2} E[N(T)] = \frac{T}{2} \cdot \lambda T = \frac{\lambda T^2}{2}. \quad \blacklozenge \end{aligned}$$

Example 12f At a winery in Napa Valley, California, starting at 10:00 A.M., tours are given every T units of time. Visitors arrive to take a tour according to a Poisson process with rate λ , independently of the times and duration of the tours. Upon arrival, visitors wait in the wine tasting room for the next tour. While waiting, they engage in wine tasting, they order hors d'oeuvres, purchase wine and wine accessories, and buy books on wine and winery guides. As a result, independently of his or her waiting time, a random visitor spends, on average, d dollars per unit of time while waiting for the next tour. Furthermore, each time a tour begins, all the visitors waiting at the wine tasting room will take the tour. What is the expected value of the total sales from the wine tasting room to the tour-taking visitors between two consecutive tours?

Solution: Label the time point 10:00 A.M. as $t = 0$. Let W be the total sale of the wine tasting room to the tour-taking visitors between 0 and T . Since the behavior of the Poisson process between any two consecutive tours is a probabilistic replica of its behavior between 0 and T , the desired quantity is $E(W)$. For $t \geq 0$, let $N(t)$ be the number of visitors who arrived at the wine tasting room at or prior to t . Given $N(T) = n$, for $1 \leq i \leq n$, let S_i be the arrival time of the i th visitor, and let Y_i be the amount in dollars, per unit of time, the i th visitor spends while waiting for the tour. We are given that, for all i , $E(Y_i) = d$. We have

$$\begin{aligned} E(W) &= \sum_{n=0}^{\infty} E[W \mid N(T) = n] P(N(T) = n) \\ &= \sum_{n=1}^{\infty} E[W \mid N(T) = n] P(N(T) = n). \end{aligned}$$

For $n \geq 1$,

$$\begin{aligned} E[W \mid N(T) = n] &= E[Y_1(T - S_1) + Y_2(T - S_2) + \cdots + Y_n(T - S_n)] \\ &= E[(Y_1 + Y_2 + \cdots + Y_n)T - (Y_1S_1 + Y_2S_2 + \cdots + Y_nS_n)] \\ &= [E(Y_1) + E(Y_2) + \cdots + E(Y_n)]T \\ &\quad - [E(Y_1)E(S_1) + E(Y_2)E(S_2) + \cdots + E(Y_n)E(S_n)] \\ &= ndT - [dE(S_1) + dE(S_2) + \cdots + dE(S_n)] \\ &= ndT - dE(S_1 + S_2 + \cdots + S_n). \end{aligned}$$

Now, let $X_1, X_2, \dots, X_n$ be n randomly selected points from the interval $[0, T]$. Let $X_{(1)}, X_{(2)}, \dots, X_{(n)}$ be the order statistics of $X_1, X_2, \dots, X_n$. Then, by Theorem 12.4 of the book,

$$\begin{aligned} E(S_1 + S_2 + \cdots + S_n) &= E(X_{(1)} + X_{(2)} + \cdots + X_{(n)}) \\ &= E(X_1 + X_1 + X_2 + \cdots + X_n) = nE(X_1) = \frac{nT}{2}. \end{aligned}$$

Hence

$$\begin{aligned}
 E(W) &= \sum_{n=1}^{\infty} E[W \mid N(T) = n] P(N(T) = n) \\
 &= \sum_{n=1}^{\infty} \left(ndT - d \frac{nT}{2} \right) \cdot \frac{e^{-\lambda T} (\lambda T)^n}{n!} = \frac{dT}{2} \sum_{n=1}^{\infty} n \frac{e^{-\lambda T} (\lambda T)^n}{n!} \\
 &= \frac{dT}{2} E[N(T)] = \frac{dT}{2} \cdot \lambda T = \frac{a\lambda T^2}{2}. \quad \blacklozenge
 \end{aligned}$$

Example 12g (Genetics) Recall that in organisms having two sets of chromosomes, called diploid organisms, which we all are, each hereditary character of each individual is carried by a pair of genes. A gene has alternate alleles which usually are dominant A or recessive a , so that the possible pairs of genes are AA , Aa (same as aA), and aa . Suppose that the zeroth generation of a diploid organism consists of *two* individuals of opposite sex of the entire population who are randomly mated. Let the first generation be *two* opposite sex offspring of the zeroth generation who are randomly mated, the second generation be *two* opposite sex offspring of the first generation who are randomly mated, and so on. For $n \geq 0$, define X_n to be $aa \times aa$ if the two individuals of the n th generation both are aa . Define X_n to be $Aa \times aa$ if one individual of the n th generation is AA and the other one is Aa . Define X_n to be $Aa \times aa$, $Aa \times Aa$, $AA \times Aa$, and $AA \times AA$ similarly. Let

$$\begin{array}{lll}
 \text{State 0} = aa \times aa & \text{State 1} = AA \times aa & \text{State 2} = Aa \times aa \\
 \text{State 3} = Aa \times Aa & \text{State 4} = AA \times Aa & \text{State 5} = AA \times AA.
 \end{array}$$

The following transition probability matrix shows that $\{X_n : n \geq 0\}$ is a Markov chain. That is, given the present state, we can determine the transition probabilities to the next step no matter what the course of transitions in the past was.

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 1/4 & 0 & 1/2 & 1/4 & 0 & 0 \\ 1/16 & 1/8 & 1/4 & 1/4 & 1/4 & 1/16 \\ 0 & 0 & 0 & 1/4 & 1/2 & 1/4 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

To see that, for example, the probabilities of one-step transitions from step 3 to various steps are given by the numbers in the fourth row, suppose that X_n is $Aa \times Aa$. Then an offspring of the n th generation is AA with probability $1/4$, Aa with probability $1/2$, and aa with probability $1/4$. Therefore, X_{n+1} is $aa \times aa$ with probability $(1/4)(1/4) = 1/16$. It is $AA \times aa$ if the father is AA and the mother is aa or vice versa. So X_{n+1} is $AA \times aa$ with probability $2(1/4)(1/4) = 1/8$. Similarly, X_{n+1} is $Aa \times aa$ with probability $2(1/2)(1/4) = 1/4$, it is $Aa \times Aa$ with probability $(1/2)(1/2) = 1/4$, $AA \times Aa$ with probability $2(1/4)(1/2) = 1/4$, and $AA \times AA$ with probability $(1/4)(1/4) = 1/16$.

It should be clear that the finite-state Markov chain above is reducible, and its communication classes are $\{0\}$, $\{1, 2, 3, 4\}$ and $\{5\}$. Clearly, states 0 and 5 are absorbing. Therefore, the classes $\{0\}$ and $\{5\}$ are positive recurrent. The class $\{1, 2, 3, 4\}$ is transient.

For the Markov chain $\{X_n: n \geq 0\}$, starting from state $i \in \{1, 2, 3, 4\}$, absorption to state 0 ($aa \times aa$) will occur at the n th generation if the state at the $(n-1)$ st generation is either 2 ($Aa \times aa$) or 3 ($Aa \times Aa$), and the next transition is into state 0. Therefore, the probability of absorption to $aa \times aa$ at the n th generation is

$$p_{i2}^{n-1}p_{20} + p_{i3}^{n-1}p_{30} = \frac{1}{4}p_{i2}^{n-1} + \frac{1}{16}p_{i3}^{n-1},$$

where p_{i2}^{n-1} and p_{i3}^{n-1} are the $i2$ -entry and $i3$ -entry of the matrix $\mathbf{P}^{n-1}$. Similarly, starting from $i \in \{1, 2, 3, 4\}$, the probability of absorption to state $AA \times AA$ at the n th generation is

$$p_{i3}^{n-1}p_{35} + p_{i4}^{n-1}p_{45} = \frac{1}{16}p_{i3}^{n-1} + \frac{1}{4}p_{i4}^{n-1}. \quad \blacklozenge$$

Example 12h (Genetics—The Wright-Fisher Model) Suppose that, in a population of N diploid organisms with alternate dominant allele A and recessive allele a , the population size remains constant for all generations. Under random mating, for $n = 0, 1, \dots$, let X_n be the number of A alleles in the n th generation. Clearly, $\{X_n: n = 0, 1, \dots\}$ is a Markov chain with state space $\{0, 1, \dots, 2N\}$ and, given $X_n = i$, X_{n+1} is a binomial random variable with parameters $2N$ and $i/2N$. (Note that there are $2N$ alleles altogether.) Hence, for $0 \leq i, j \leq 2N$, the transition probabilities are given by

$$p_{ij} = \binom{2N}{j} \left(\frac{i}{2N}\right)^j \left(1 - \frac{i}{2N}\right)^{2N-j}.$$

Furthermore,

$$E(X_{n+1} | X_n = i) = 2N \cdot \frac{i}{2N} = i.$$

Hence

$$E(X_{n+1} | X_n) = X_n.$$

This implies that

$$E(X_{n+1}) = E[E(X_{n+1} | X_n)] = E(X_n),$$

showing that the expected value of the A alleles is the same for all generations.

The Markov chain $\{X_n: n = 0, 1, \dots\}$ has two absorbing states, 0 and $2N$. Absorbing to 0 means permanent loss of allele A from the population. Obviously, if for some generation, all individuals are aa , then, from that generation on, the individuals of all future generations will also be aa . Similarly, absorbing to $2N$ means permanent loss of allele a from the population and, hence, all of its future generations. Let x_i be the probability that the Markov chain will be absorbed into 0 or $2N$ given that, initially, the number of A alleles is i . By the law of total probability,

$$\begin{cases} x_i = \sum_{j=0}^{2N} \binom{2N}{j} \left(\frac{i}{2N}\right)^j \left(1 - \frac{i}{2N}\right)^{2N-j} x_j, & 1 \leq i \leq 2N-1, \\ x_0 = x_{2N} = 1. \end{cases}$$

It is readily seen that the solution to this system of $2N + 1$ equations in $2N + 1$ unknowns is

$$x_0 = x_1 = \cdots = x_{2N} = 1.$$

This is true since by the binomial expansion,

$$\sum_{j=0}^{2N} \binom{2N}{j} \left(\frac{i}{2N}\right)^j \left(1 - \frac{i}{2N}\right)^{2N-j} = \left(\frac{i}{2N} + 1 - \frac{i}{2N}\right)^{2N} = 1.$$

Having $x_0 = x_1 = \cdots = x_{2N} = 1$ shows that no matter what the number of A alleles in the initial population is, the Markov chain will eventually be absorbed into state 0 or state $2N$ with probability 1. We will now find y_i , the probability that the Markov chain will be absorbed into state $2N$ given that $X_0 = i$, $0 \leq i \leq 2N$. Note that $(y_0, y_1, \dots, y_{2N})$ is the solution to the following system of linear equations obtained by the law of total probability:

$$\begin{cases} y_0 = 0 \\ y_i = \sum_{j=0}^{2N} \binom{2N}{j} \left(\frac{i}{2N}\right)^j \left(1 - \frac{i}{2N}\right)^{2N-j} y_j, & 1 \leq i \leq 2N - 1 \\ y_{2N} = 1. \end{cases}$$

If we show that, for $0 \leq i \leq N$, $y_i = i/2N$ satisfy the equations of this system, we are done. The fact that, for $i = 0$ and $2N$, $y_i = i/2N$ is trivial. For $1 \leq i \leq 2N - 1$,

$$\begin{aligned} \sum_{j=0}^{2N} \binom{2N}{j} \left(\frac{i}{2N}\right)^j \left(1 - \frac{i}{2N}\right)^{2N-j} y_j &= \sum_{j=0}^{2N} \binom{2N}{j} \left(\frac{i}{2N}\right)^j \left(1 - \frac{i}{2N}\right)^{2N-j} \cdot \frac{j}{2N} \\ &= \frac{1}{2N} \sum_{j=0}^{2N} j \binom{2N}{j} \left(\frac{i}{2N}\right)^j \left(1 - \frac{i}{2N}\right)^{2N-j} \\ &= \frac{1}{2N} \cdot 2N \cdot \frac{i}{2N} = \frac{i}{2N} = y_i, \end{aligned}$$

where

$$\sum_{j=0}^{2N} j \binom{2N}{j} \left(\frac{i}{2N}\right)^j \left(1 - \frac{i}{2N}\right)^{2N-j} = 2N \cdot \frac{i}{2N}$$

follows since it is the expected value of a binomial random variable with parameters $2N$ and $i/2N$. Hence, if initially the number of A alleles is i , the probability that eventually all of the individuals in the population are AA is $i/2N$. The probability that eventually they are all aa is $1 - (i/2N)$. ♦

12B BROWNIAN MOTION

To show that all material is made up from molecules, in 1827, the English Botanist Robert Brown (1773–1851) studied the motion of pollen particles in a container of water. He observed that, even though liquid in a container may appear to be motionless, particles suspended in the liquid, under constant and incessant random collisions with nearby particles, undergo unceasing motions in a totally erratic way. In 1905, unaware of Robert Brown's work, Albert Einstein (1879–1955) presented the first mathematical description of this phenomenon using laws of physics. After Einstein, much work was done by many scientists to advance the physical theory of the motion of particles in liquid. However, it was in 1923 that Norbert Wiener (1894–1964) was able to formulate Robert Brown's observations with mathematical rigor.

Suppose that liquid in a cubic container is placed in a coordinate system, and at time 0, a particle is at $(0, 0, 0)$, the origin. Unless otherwise specified, for mathematical simplicity, we assume that the container is unbounded from all sides and is full of liquid. Let $(X(t), Y(t), Z(t))$ be the position of the particle after t units of time. We will find the distribution functions of $X(t)$, $Y(t)$, and $Z(t)$. To do so, we will find the distribution function of $X(t)$. It should be clear that the distribution functions of $Y(t)$ and $Z(t)$ are similarly calculated and are identical to that of the distribution function of $X(t)$. Observe that, in infinitesimal lengths of times during the time period $(0, t)$, $X(t)$ is a sum of infinitely many small movements in the direction of the x -coordinate. One way to find it is to divide the time interval t into $n = [t/h]$ subintervals, where the length of each subinterval is h , for some infinitesimal h , and $[t/h]$ is the greatest integer less than or equal to t/h . Suppose that, for infinitesimal $\delta > 0$, in each of these time subintervals, the x -coordinate of the particle moves to the right δ units with probability $1/3$, moves to the left δ units with probability $1/3$, and does not move with probability $1/3$. For $i \geq 1$, let

$$X_i = \begin{cases} \delta & \text{with probability } 1/3 \\ -\delta & \text{with probability } 1/3 \\ 0 & \text{with probability } 1/3. \end{cases}$$

Then $\{X_i: i \geq 1\}$ is a random walk with $E(X_i) = 0$ and

$$\text{Var}(X_i) = E(X_i^2) = \frac{1}{3} \cdot \delta^2 + \frac{1}{3} \cdot (-\delta)^2 + \frac{1}{3} \cdot 0^2 = \frac{2\delta^2}{3}.$$

For large n , it should be clear that $X(t)$ and $\sum_{i=1}^n X_i$ have approximately the same distribution. Since the motions of the particle are totally erratic, $X_1, X_2, \dots$ are independent random variables. Hence, by the central limit theorem, for large n , the distribution of $\sum_{i=1}^n X_i$ is approximately $N\left(0, \frac{2\delta^2}{3}n\right)$ or $N\left(0, \frac{2\delta^2}{3h}t\right)$. For the limiting behavior of the random walk $\{X_i: i \geq 1\}$ to closely approximate the x -coordinate, $X(t)$, of the position of the particle after t units of time, we need to let $h \rightarrow 0$ and $\delta \rightarrow 0$ in a way that $(2\delta^2)/(3h)$ remains a constant σ^2 . Doing this, we obtain that, for all $t > 0$, $X(t)$ is a normal random variable with mean 0 and variance $\sigma^2 t$. The same argument can be used to show that $Y(t)$, the y -coordinate, and $Z(t)$, the z -coordinate of the position of the particle after t units of time, are also normal with mean 0 and variance $\sigma^2 t$. It can be shown that

$X(t)$, $Y(t)$, and $Z(t)$ are independent random variables.

Moreover, physical observations show that they possess stationary and independent increments. For the process $\{X(t): t \geq 0\}$, by **stationary increments** we mean that, for $s < t$ and $h \in (-\infty, \infty)$, the random variables $X(t) - X(s)$ and $X(t+h) - X(s+h)$ are identically distributed. Similarly, as previously defined, we say that $\{X(t): t \geq 0\}$ possesses independent increments if for $s \leq t \leq u \leq v$, we have that $X(t) - X(s)$ and $X(v) - X(u)$ are independent random variables. The properties of the processes $\{X(t): t \geq 0\}$, $\{Y(t): t \geq 0\}$ and $\{Z(t): t \geq 0\}$ obtained by studying the x , y , and z coordinates of the positions of a particles's motion in a liquid motivate the following definition.

Definition 12a A stochastic process $\{X(t): t \geq 0\}$ with state space $\mathcal{S} = (-\infty, \infty)$ that possesses stationary and independent increments is said to be a **Brownian motion** if $X(0) = 0$ and, for $t > 0$, $X(t)$ is a normal random variable with mean 0 and variance $\sigma^2 t$, for some $\sigma > 0$.

Brownian motions are also called **Wiener processes**. σ^2 is called the **variance parameter** of the Brownian motion. For $t > 0$, the probability density function of $X(t)$ is denoted by $\phi_t(x)$. Therefore,

$$\phi_t(x) = \frac{1}{\sigma\sqrt{2\pi t}} \exp \left[-\frac{x^2}{2\sigma^2 t} \right]. \quad (1)$$

For a Brownian motion $\{X(t): t \geq 0\}$, note that by stationarity of the increments:

$$\text{For } s, t \geq 0, X(t+s) - X(s) \text{ is } N(0, \sigma^2 t).$$

Brownian motion is an important area in stochastic processes. It has diverse applications in various fields. Even though its origin is in the studies of motions of particles in liquid or gas, it is used in quantum mechanics, stock market fluctuations, and testing of goodness of fit in statistics.

Let $\{X(t): t \geq 0\}$ be a Brownian motion with variance parameter σ^2 . Let $W(t) = X(t)/\sigma$. It is straightforward to show that $\{W(t): t \geq 0\}$ is a Brownian motion with variance parameter 1. A Brownian motion with variance parameter 1 is called a **standard Brownian motion**. For simplicity, we sometimes prove theorems about a Brownian motion with variance parameter 1 and then use the transformation $W(t) = X(t)/\sigma$ to find the more general theorem.

Intuitively, it is not difficult to see that the graph of the x -coordinates of the positions of a particle's motion in a liquid, as a function of t , is continuous. This is true for the y - and z -coordinates as well. It is interesting to know that these functions are *everywhere continuous but nowhere differentiable*. Therefore, being at no point smooth, they are extremely kinky. The study of nowhere differentiability of such graphs is difficult and beyond the scope of this book. However, from a mechanical point of view, nowhere differentiability is a result of the assumption that the mass of the particle in motion is 0.

We will now discuss, step by step, some of the most important properties of Brownian motions.

The Conditional Probability Density Function of $X(t)$ Given that $X(0) = x_0$

In Section 12.3 of the book, we showed that for a discrete-time Markov chain, p_{ij}^n , the probability of moving from state i to state j in n steps satisfies the Chapman-Kolmogorov equations. In Section 12.4 of the book, we proved that $p_{ij}(t)$, the probability that a continuous-time Markov chain moves from state i to state j in t units of time satisfies the Kolmogorov forward and backward equations. To point out the parallel between these and analogous results in Brownian motions, let $f_{t|0}(x|x_0)$ be the conditional probability density function of $X(t)$ given that $X(0) = x_0$.[†] This is the function analogous to $p_{ij}(t)$ for continuous-time Markov chain, and p_{ij}^n for the discrete-time Markov chain. The functions $f_{t|0}(x|x_0)$ are called **transition probability density functions** of the Brownian motion. They should not be confused with $f_{X|Y}(x|y)$, the conditional probability density function of a random variable X given that $Y = y$.

Note that, by definition of a probability density function, for $u \in (-\infty, \infty)$,

$$P(X(t) \leq u \mid X(0) = x_0) = \int_{-\infty}^u f_{t|0}(x|x_0) dx.$$

Also note that, since Brownian motions possess stationary increments, the probability density function of $X(t + t_0)$ given that $X(t_0) = x_0$ is $f_{t|0}(x|x_0)$ as well. By selecting appropriate scales, Albert Einstein showed that $f_{t|0}(x|x_0)$ satisfies the partial differential equation

$$\frac{\partial f}{\partial t} = \frac{1}{2}\sigma^2 \frac{\partial^2 f}{\partial x^2}, \quad (2)$$

called the **backward diffusion equation**. Considering the facts that $f_{t|0}(x|x_0) \geq 0$, $\int_{-\infty}^{\infty} f_{t|0}(x|x_0) dx = 1$, and $\lim_{t \rightarrow 0} f_{t|0}(x|x_0) = 0$ if $x \neq x_0$, it is straightforward to see that the unique solution to the backward diffusion equation under these conditions is

$$f_{t|0}(x|x_0) = \frac{1}{\sigma\sqrt{2\pi t}} \exp\left[-\frac{(x - x_0)^2}{2\sigma^2 t}\right], \quad (3)$$

which is the probability density function of a normal random variable with mean x_0 and standard deviation $\sigma\sqrt{t}$.

The Joint Probability Density Function of $X(t_1), X(t_2), \dots, X(t_n)$

Next, for $t_1 < t_2$, let $f(x_1, x_2)$ be the joint probability density function of $X(t_1)$ and $X(t_2)$. To find this important function, let $U = X(t_1)$ and $V = X(t_2) - X(t_1)$. Applying Theorem 8.8 of the book, we will first find the joint probability density function of U and V . Observe that the system of equations

$$\begin{cases} x_1 = u \\ x_2 - x_1 = v \end{cases}$$

[†]Even though, in general, we adopt the convention of $X(0) = 0$, this choice is not necessary. It is not contradictory to assume that $X(0) = x_0$, for some point $x_0 \neq 0$.

has the unique solution $x_1 = u$, $x_2 = u + v$. Since

$$\mathbf{J} = \begin{vmatrix} \frac{\partial x_1}{\partial u} & \frac{\partial x_1}{\partial v} \\ \frac{\partial x_2}{\partial u} & \frac{\partial x_2}{\partial v} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = 1 \neq 0,$$

we have that, $g(u, v)$, the joint probability density function of U and V is given by

$$g(u, v) = f(u, u + v)|\mathbf{J}| = f(u, u + v).$$

By this relation, $f(x_1, x_2) = g(x_1, x_2 - x_1)$. Since U and V are independent random variables, $g(x_1, x_2 - x_1)$ is the product of the marginal probability density functions of $X(t_1)$ and $X(t_2) - X(t_1)$, which by the stationarity and independence of the increments of Brownian motion processes, are $N(0, \sigma^2 t_1)$ and $N(0, \sigma^2(t_2 - t_1))$, respectively. Hence

$$f(x_1, x_2) = \frac{1}{\sigma\sqrt{2\pi t_1}} \exp\left(-\frac{x_1^2}{2\sigma^2 t_1}\right) \cdot \frac{1}{\sigma\sqrt{2\pi(t_2 - t_1)}} \exp\left[-\frac{(x_2 - x_1)^2}{2\sigma^2(t_2 - t_1)}\right],$$

or, equivalently,

$$f(x_1, x_2) = \frac{1}{2\sigma^2\pi\sqrt{t_1(t_2 - t_1)}} \exp\left[-\frac{1}{2\sigma^2}\left(\frac{x_1^2}{t_1} + \frac{(x_2 - x_1)^2}{t_2 - t_1}\right)\right], \quad -\infty < x_1, x_2 < \infty. \quad (4)$$

For $t_1 < t_2 < \dots < t_n$, let $f(x_1, x_2, \dots, x_n)$ be the joint probability density function of $X(t_1), X(t_2), \dots, X(t_n)$. Let $g(x_1, x_2, \dots, x_n)$ be the joint probability density function of $X(t_1), X(t_2) - X(t_1), \dots, X(t_n) - X(t_{n-1})$. An argument similar to the one presented for the case $n = 2$, shows that

$$f(x_1, x_2, \dots, x_n) = g(x_1, x_2 - x_1, \dots, x_n - x_{n-1}).$$

Since the random variables $X(t_1), X(t_2) - X(t_1), \dots, X(t_n) - X(t_{n-1})$ are independent, $X(t_1) \sim N(0, \sigma^2 t_1)$ and, for $2 \leq i \leq n$, $X(t_i) - X(t_{i-1}) \sim N(0, \sigma^2(t_i - t_{i-1}))$. Thus

$$f(x_1, x_2, \dots, x_n) = \frac{\exp\left[-\frac{1}{2\sigma^2}\left(\frac{x_1^2}{t_1} + \sum_{i=1}^{n-1} \frac{(x_{i+1} - x_i)^2}{t_{i+1} - t_i}\right)\right]}{\sigma^n \sqrt{(2\pi)^n t_1(t_2 - t_1) \cdots (t_n - t_{n-1})}}, \quad -\infty < x_1, x_2, \dots, x_n < \infty. \quad (5)$$

For $t_1 < t < t_2$, the Conditional Probability Density Function of $X(t)$ Given That $X(t_1) = x_1$ and $X(t_2) = x_2$

Let us begin with the special case in which $t_1 = 0$, $x_1 = 0$, $t_2 = u_1$, and $x_2 = 0$. In such a case, the x -coordinate of the particle under consideration in the liquid, which at time 0 was at 0, is again at 0 at time u_1 . For $0 < u < u_1$, we want to find the conditional probability density function of $X(u)$ given that $X(u_1) = 0$. [The condition $X(0) = 0$ is automatically assumed for all Brownian motions unless otherwise stated.] Let $f(x, y)$ be the joint probability density function of $X(u)$ and $X(u_1)$. Then, by (4),

$$f(x, y) = \frac{1}{2\sigma^2\pi\sqrt{u(u_1-u)}} \exp\left[-\frac{1}{2\sigma^2}\left(\frac{x^2}{u} + \frac{(y-x)^2}{u_1-u}\right)\right], \quad -\infty < x, y < \infty. \quad (6)$$

Let $f_{X(u)|X(u_1)}(x|0)$ be the conditional probability density function of $X(u)$ given that $X(u_1) = 0$. Then, by relation (8.21) of the book and (1),

$$f_{X(u)|X(u_1)}(x|0) = \frac{f(x, 0)}{f_{X(u_1)}(0)} = \frac{f(x, 0)}{\phi_{u_1}(0)} = \sigma\sqrt{2\pi u_1}f(x, 0).$$

Thus, by (6),

$$\begin{aligned} f_{X(u)|X(u_1)}(x|0) &= \sigma\sqrt{2\pi u_1} \cdot \frac{1}{2\sigma^2\pi\sqrt{u(u_1-u)}} \exp\left[-\frac{1}{2\sigma^2}\left(\frac{x^2}{u} + \frac{x^2}{u_1-u}\right)\right] \\ &= \frac{1}{\sigma\sqrt{2\pi}} \sqrt{\frac{u_1}{u(u_1-u)}} \cdot \exp\left[-\frac{1}{2\sigma^2} \cdot \frac{u_1}{u(u_1-u)} x^2\right], \quad -\infty < x < \infty, \end{aligned}$$

which is the probability density function of a normal random variable with mean 0 and variance $\frac{\sigma^2 u(u_1-u)}{u_1}$. Therefore, for $0 < u < u_1$,

$$E[X(u) \mid X(0) = 0 \text{ and } X(u_1) = 0] = 0 \quad (7)$$

and

$$\text{Var}[X(u) \mid X(0) = 0 \text{ and } X(u_1) = 0] = \frac{\sigma^2 u(u_1-u)}{u_1}. \quad (8)$$

Now, for $t_1 < t < t_2$, to find the conditional probability density function of $X(t)$ given that $X(t_1) = x_1$ and $X(t_2) = x_2$, define

$$\tilde{X}(t) = X(t+t_1) - x_1 - \frac{t}{t_2-t_1}(x_2-x_1).$$

This is equivalent to

$$\tilde{X}(t-t_1) = X(t) - x_1 - \frac{t-t_1}{t_2-t_1}(x_2-x_1). \quad (9)$$

Clearly, for $t = t_1$, we obtain

$$\tilde{X}(0) = X(t_1) - x_1 = 0,$$

and

$$\tilde{X}(t_2 - t_1) = X(t_2) - x_1 - (x_2 - x_1) = 0.$$

Since $t_1 < t < t_2$ implies that $0 < t - t_1 < t_2 - t_1$, in (7) and (8) letting $u_1 = t_2 - t_1$ and $u = t - t_1$, for the process $\tilde{X}(t)$, we have

$$E[\tilde{X}(t - t_1) \mid \tilde{X}(0) = 0 \text{ and } \tilde{X}(t_2 - t_1) = 0] = 0$$

and

$$\begin{aligned} \text{Var}[\tilde{X}(t - t_1) \mid \tilde{X}(0) = 0 \text{ and } \tilde{X}(t_2 - t_1) = 0] \\ = \frac{\sigma^2(t - t_1)[(t_2 - t_1) - (t - t_1)]}{t_2 - t_1} = \frac{\sigma^2(t_2 - t)(t - t_1)}{t_2 - t_1}. \end{aligned}$$

Now, by (9),

$$\begin{aligned} 0 &= E[\tilde{X}(t - t_1) \mid \tilde{X}(0) = 0 \text{ and } \tilde{X}(t_2 - t_1) = 0] \\ &= E[X(t) \mid X(t_1) = x_1 \text{ and } X(t_2) = x_2] - x_1 - \frac{t - t_1}{t_2 - t_1}(x_2 - x_1). \end{aligned}$$

Hence

$$E[X(t) \mid X(t_1) = x_1 \text{ and } X(t_2) = x_2] = x_1 + \frac{x_2 - x_1}{t_2 - t_1}(t - t_1),$$

which is the equation of a line passing through (t_1, x_1) and (t_2, x_2) . Similarly, by (9),

$$\begin{aligned} \text{Var}[X(t) \mid X(t_1) = x_1 \text{ and } X(t_2) = x_2] \\ = \text{Var}[\tilde{X}(t - t_1) \mid \tilde{X}(0) = 0 \text{ and } \tilde{X}(t_2 - t_1) = 0] = \sigma^2 \frac{(t_2 - t)(t - t_1)}{t_2 - t_1}. \end{aligned}$$

We have shown the following important theorem:

Theorem 12a For $t_1 < t < t_2$, the conditional probability density function of $X(t)$ given that $X(t_1) = x_1$ and $X(t_2) = x_2$ is normal with mean $x_1 + \frac{x_2 - x_1}{t_2 - t_1}(t - t_1)$ and variance $\sigma^2 \frac{(t_2 - t)(t - t_1)}{t_2 - t_1}$.

Example 12i Suppose that liquid in a container is placed in a coordinate system, and at time 0, a pollen particle suspended in the liquid is at $(0, 0, 0)$, the origin. Let $X(t)$ be the x -coordinate of the position of the pollen after t minutes. Suppose that $\{X(t) : t \geq 0\}$ is a Brownian motion with variance parameter 4 and, after one minute, the x -coordinate of the pollen's position is 2.

- (a) What is the probability that after 2 minutes it is between 0 and 1?
- (b) What is the expected value and variance of the x -coordinate of the position of pollen after 30 seconds?

Solution:

(a) The desired probability is

$$\begin{aligned} P(0 < X(2) < 1 \mid X(1) = 2) &= P(-2 < X(2) - X(1) < -1 \mid X(1) = 2) \\ &= P(-2 < X(2) - X(1) < -1), \end{aligned}$$

by the independent-increments property of Brownian motion. Since $X(2) - X(1)$ is normal with mean 0 and variance $(2 - 1)\sigma^2 = 4$, letting $Z \sim N(0, 1)$, we have

$$\begin{aligned} P(-2 < X(2) - X(1) < -1) &= P\left(\frac{-2-0}{2} < Z < \frac{-1-0}{2}\right) \\ &= P(-1 < Z < -0.5) \\ &= \Phi(-0.5) - \Phi(-1) = 0.1523. \end{aligned}$$

(b) In Theorem 12a, let $t_1 = 0$, $t_2 = 1$, $x_1 = 0$, $x_2 = 2$. We have

$$\begin{aligned} E[X(1/2) \mid X(0) = 0 \text{ and } X(1) = 2] &= 0 + \frac{2-0}{1-0}\left(\frac{1}{2} - 0\right) = 1 \\ \text{Var}[X(1/2) \mid X(0) = 0 \text{ and } X(1) = 2] &= 4 \cdot \frac{\left(1 - \frac{1}{2}\right)\left(\frac{1}{2} - 0\right)}{1-0} = 1. \quad \blacklozenge \end{aligned}$$

First Passage Time Distribution

Let $\{X(t) : t \geq 0\}$ be a Brownian motion with variance parameter σ^2 . The **first passage time** to a point α , also called the **time of hitting α first**, denoted by T_α , is the minimum of the set $\{t : X(t) = \alpha\}$. We will first find $P(T_\alpha \leq t)$, $-\infty < t < \infty$, for $\alpha > 0$. Then we will calculate it for $\alpha < 0$. For $\alpha > 0$, note that, by Theorem 3.4 of the book,

$$\begin{aligned} P(X(t) \geq \alpha) &= P(X(t) \geq \alpha \mid T_\alpha < t)P(T_\alpha < t) + P(X(t) \geq \alpha \mid T_\alpha = t)P(T_\alpha = t) \\ &\quad + P(X(t) \geq \alpha \mid T_\alpha > t)P(T_\alpha > t). \end{aligned} \quad (10)$$

Since $\alpha > 0$, and the graph of $X(t)$, as a function of t , is continuous, it follows that $P(X(t) \geq \alpha \mid T_\alpha > t) = 0$. Since T_α is a continuous random variable, $P(T_\alpha = t) = 0$. Observe that after the process hits α , for $t > T_\alpha$, by symmetry, it is equally likely that $X(t) \geq \alpha$ and $X(t) \leq \alpha$. Hence

$$P(X(t) \geq \alpha \mid T_\alpha < t) = \frac{1}{2}.$$

Incorporating these facts into (10), we obtain

$$P(X(t) \geq \alpha) = \frac{1}{2}P(T_\alpha < t).$$

Since $X(t) \sim N(0, \sigma^2 t)$, we have

$$P(T_\alpha \leq t) = P(T_\alpha < t) = 2P(X(t) \geq \alpha) = \frac{2}{\sigma\sqrt{2\pi t}} \int_\alpha^\infty e^{-x^2/(2t\sigma^2)} dx.$$

Letting $\frac{x}{\sigma\sqrt{t}} = y$, this reduces to

$$P(T_\alpha \leq t) = \frac{2}{\sqrt{2\pi}} \int_{\alpha/(\sigma\sqrt{t})}^{\infty} e^{-y^2/2} dy = 2 \left[1 - \Phi\left(\frac{\alpha}{\sigma\sqrt{t}}\right) \right]. \quad (11)$$

To find $P(T_\alpha \leq t)$ for $\alpha < 0$, note that T_α and $T_{-\alpha}$ are identically distributed. Hence

$$P(T_\alpha \leq t) = P(T_{-\alpha} \leq t) = 2 \left[1 - \Phi\left(-\frac{\alpha}{\sigma\sqrt{t}}\right) \right]. \quad (12)$$

Putting (11) and (12) together, we have that for all $\alpha \in (-\infty, \infty)$, the distribution function of the first passage time to α is given by

$$P(T_\alpha \leq t) = 2 \left[1 - \Phi\left(\frac{|\alpha|}{\sigma\sqrt{t}}\right) \right]. \quad (13)$$

As in the case of symmetric random walks, it can be shown that, even though, with probability 1, $\{X(t) : t \geq 0\}$ eventually will hit α , none of the moments of the first passage time to α is finite.

The Maximum of a Brownian Motion

As before, let T_α be the first passage time to α . By the continuity of the paths of Brownian motions, for $\alpha > 0$, the event $\max_{0 \leq s \leq t} X(s) \geq \alpha$ occurs if and only if $T_\alpha \leq t$. Therefore,

$$P\left(\max_{0 \leq s \leq t} X(s) \geq \alpha\right) = P(T_\alpha \leq t) = 2 \left[1 - \Phi\left(\frac{\alpha}{\sigma\sqrt{t}}\right) \right],$$

or, equivalently,

$$P\left(\max_{0 \leq s \leq t} X(s) \leq \alpha\right) = 1 - 2 \left[1 - \Phi\left(\frac{\alpha}{\sigma\sqrt{t}}\right) \right] = 2\Phi\left(\frac{\alpha}{\sigma\sqrt{t}}\right) - 1.$$

Thus the distribution function of $\max_{0 \leq s \leq t} X(s)$ is given by

$$P\left(\max_{0 \leq s \leq t} X(s) \leq x\right) = \begin{cases} 2\Phi\left(\frac{x}{\sigma\sqrt{t}}\right) - 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0. \end{cases} \quad (14)$$

The Zeros of Brownian Motion

Let $\{X(t) : t \geq 0\}$ be a Brownian motion with variance parameter σ^2 . In this subsection, we will find the probability that $X(t) = 0$ at least once in the time interval (t_1, t_2) , $0 < t_1 < t_2$. The event that $X(t) = 0$ for at least one t in this interval is

$$\bigcup_{t_1 < t < t_2} \{X(t) = 0\}.$$

To begin with, note that if $X(t_1) = a > 0$, then, by the continuity of the Brownian motion's paths, $\bigcup_{t_1 < t < t_2} \{X(t) = 0\}$ occurs if and only if $\min_{t_1 < t < t_2} X(t) \leq 0$. Hence, for $a > 0$,

$$P\left(\bigcup_{t_1 < t < t_2} \{X(t) = 0\} \mid X(t_1) = a\right) = P\left(\min_{t_1 < t < t_2} X(t) \leq 0 \mid X(t_1) = a\right). \quad (15)$$

For $a < 0$, we consider the Brownian motion $\{-X(t) : t \geq 0\}$. Since, for all $t \geq 0$, $X(t)$ and $-X(t)$ are identically distributed, we have

$$\begin{aligned} P\left(\bigcup_{t_1 < t < t_2} \{X(t) = 0\} \mid X(t_1) = a\right) &= P\left(\bigcup_{t_1 < t < t_2} \{-X(t) = 0\} \mid -X(t_1) = a\right) \\ &= P\left(\bigcup_{t_1 < t < t_2} \{X(t) = 0\} \mid X(t_1) = -a\right), \end{aligned}$$

which, by (15), is equal to $P\left(\min_{t_1 < t < t_2} X(t) \leq 0 \mid X(t_1) = -a\right)$ since $-a > 0$. This and (15) imply that, for all $a \in (-\infty, \infty)$,

$$P\left(\bigcup_{t_1 < t < t_2} \{X(t) = 0\} \mid X(t_1) = a\right) = P\left(\min_{t_1 < t < t_2} X(t) \leq 0 \mid X(t_1) = |a|\right). \quad (16)$$

Now, again, using the fact that $X(t)$ and $-X(t)$ are identically distributed, and

$$-\min_{t_1 < t < t_2} -X(t) = \max_{t_1 < t < t_2} X(t),$$

we have

$$\begin{aligned} P\left(\min_{t_1 < t < t_2} X(t) \leq 0 \mid X(t_1) = |a|\right) &= 1 - P\left(\min_{t_1 < t < t_2} X(t) > 0 \mid X(t_1) = |a|\right) \\ &= 1 - P\left(\min_{t_1 < t < t_2} -X(t) > 0 \mid -X(t_1) = |a|\right) \\ &= 1 - P\left(-\min_{t_1 < t < t_2} -X(t) < 0 \mid -X(t_1) = |a|\right) \\ &= 1 - P\left(\max_{t_1 < t < t_2} X(t) < 0 \mid X(t_1) = -|a|\right) \\ &= P\left(\max_{t_1 < t < t_2} X(t) \geq 0 \mid X(t_1) = -|a|\right) \\ &= P\left(\max_{t_1 < t < t_2} X(t) \geq |a| \mid X(t_1) = 0\right) \\ &= P\left(\max_{0 < t < t_2 - t_1} X(t) \geq |a| \mid X(0) = 0\right) \\ &= 2\left[1 - \Phi\left(\frac{|a|}{\sigma\sqrt{t_2 - t_1}}\right)\right], \end{aligned} \quad (17)$$

where the last equality follows from (14). Now, on the one hand, by conditioning on $X(t_1)$

and by relations (16) and (17), we obtain

$$\begin{aligned}
 P\left(\bigcup_{t_1 < t < t_2} \{X(t) = 0\}\right) &= \int_{-\infty}^{\infty} P\left(\bigcup_{t_1 < t < t_2} \{X(t) = 0\} \mid X(t_1) = a\right) \phi_{t_1}(a) da \\
 &= \int_{-\infty}^{\infty} P\left(\min_{t_1 < t < t_2} X(t) \leq 0 \mid X(t_1) = |a|\right) \phi_{t_1}(a) da \\
 &= \int_{-\infty}^{\infty} 2\left[1 - \Phi\left(\frac{|a|}{\sigma\sqrt{t_2 - t_1}}\right)\right] \phi_{t_1}(a) da. \tag{18}
 \end{aligned}$$

On the other hand, letting $Z \sim N(0, 1)$, we get

$$\begin{aligned}
 P(X(t_2) - X(t_1) > |X(t_1)|) &= \int_{-\infty}^{\infty} P(X(t_2) - X(t_1) > |X(t_1)| \mid X(t_1) = a) \phi_{t_1}(a) da \\
 &= \int_{-\infty}^{\infty} P(X(t_2) - X(t_1) > |a|) \phi_{t_1}(a) da \\
 &= \int_{-\infty}^{\infty} P\left(Z > \frac{|a|}{\sigma\sqrt{t_2 - t_1}}\right) \phi_{t_1}(a) da \\
 &= \int_{-\infty}^{\infty} \left[1 - \Phi\left(\frac{|a|}{\sigma\sqrt{t_2 - t_1}}\right)\right] \phi_{t_1}(a) da. \tag{19}
 \end{aligned}$$

Comparing (18) and (19) yields

$$P\left(\bigcup_{t_1 < t < t_2} \{X(t) = 0\}\right) = 2P(X(t_2) - X(t_1) > |X(t_1)|).$$

Let Z_1 and Z_2 be two independent standard normal random variables. Then $X(t_2) - X(t_1)$ is identically distributed with $\sigma\sqrt{t_2 - t_1}Z_1$ and $X(t_1)$ is identically distributed with $\sigma\sqrt{t_1}Z_2$. Hence

$$P\left(\bigcup_{t_1 < t < t_2} \{X(t) = 0\}\right) = 2P(\sigma\sqrt{t_2 - t_1}Z_1 > |\sigma\sqrt{t_1}Z_2|) = 2P\left(\frac{Z_1}{|Z_2|} > \sqrt{\frac{t_1}{t_2 - t_1}}\right).$$

By Example 10.24 of the book, $Z_1/|Z_2|$ is a Cauchy random variable. Therefore,

$$\begin{aligned}
 P\left(\bigcup_{t_1 < t < t_2} \{X(t) = 0\}\right) &= 2 \int_{\sqrt{t_1/(t_2 - t_1)}}^{\infty} \frac{1}{\pi(1 + x^2)} dx \\
 &= \frac{2}{\pi} \left[\arctan x \right]_{\sqrt{t_1/(t_2 - t_1)}}^{\infty} = \frac{2}{\pi} \left[\frac{\pi}{2} - \arctan \sqrt{\frac{t_1}{t_2 - t_1}} \right] \\
 &= 1 - \frac{2}{\pi} \arctan \sqrt{\frac{t_1}{t_2 - t_1}}. \tag{20}
 \end{aligned}$$

Now draw a right triangle, such as the one in Figure 12a, with hypotenuse of length $\sqrt{t_2}$ and sides of lengths $\sqrt{t_1}$ and $\sqrt{t_2 - t_1}$. Let the angle adjacent to the side of length $\sqrt{t_1}$ be θ radians. Then

$$\tan\left(\frac{\pi}{2} - \theta\right) = \frac{\sqrt{t_1}}{\sqrt{t_2 - t_1}},$$

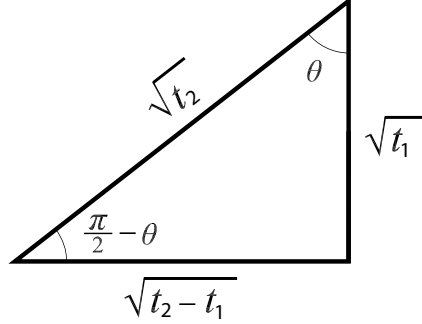


Figure 12a Triangle of Theorem 12b.

which implies that

$$\arctan \sqrt{\frac{t_1}{t_2 - t_1}} = \frac{\pi}{2} - \theta,$$

or

$$1 - \frac{2}{\pi} \arctan \sqrt{\frac{t_1}{t_2 - t_1}} = \frac{2}{\pi} \theta = \frac{2}{\pi} \arccos \sqrt{\frac{t_1}{t_2}}.$$

Considering this identity and (20), we have shown the following theorem:

Theorem 12b For a Brownian motion $\{X(t) : t \geq 0\}$ with variance parameter σ^2 , the probability that $X(t) = 0$ at least once in the time interval (t_1, t_2) , $0 < t_1 < t_2$, is $\frac{2}{\pi} \arccos \sqrt{\frac{t_1}{t_2}}$.

Brownian Motion with Drift

Let $\{W(t) : t \geq 0\}$ be a standard Brownian motion. That is, a Brownian motion with variance parameter 1. For $\mu \in (-\infty, \infty)$ and $\sigma > 0$, the process

$$X(t) = \mu t + \sigma W(t), \quad t \geq 0,$$

is called the **Brownian motion with drift parameter μ and variance parameter σ^2** . Note that for such a process, it is still the case that $X(0) = 0$ and $\{X(t) : t \geq 0\}$ possesses stationary and independent increments. However, for $t > 0$, $X(t)$ is normal with mean μt and variance $\sigma^2 t$.

For a Brownian motion $\{X(t) : t \geq 0\}$ with drift parameter μ and variance parameter σ^2 , let $f_{t|0}(x|x_0)$ be the conditional probability density function of $X(t)$ given that $X(0) = x_0$, or, equivalently, the conditional probability density function of $X(t + t_0)$ given that $X(t_0) = x_0$. To find $f_{t|0}(x|x_0)$, note that

$$\begin{aligned} P(X(t) \leq x \mid X(0) = x_0) &= P(\mu t + \sigma W(t) \leq x \mid \sigma W(0) = x_0) \\ &= P\left(W(t) \leq \frac{x - \mu t}{\sigma} \mid W(0) = \frac{x_0}{\sigma}\right) \\ &= P\left(W(t) - W(0) \leq \frac{x - x_0 - \mu t}{\sigma} \mid W(0) = \frac{x_0}{\sigma}\right). \end{aligned}$$

Let $Z \sim N(0, 1)$; since $W(t) - W(0)$ and $W(0)$ are independent random variables, and $W(t) - W(0)$ is $N(0, t)$, this equation gives

$$\begin{aligned} P(X(t) \leq x \mid X(0) = x_0) &= P\left(W(t) - W(0) \leq \frac{x - x_0 - \mu t}{\sigma}\right) \\ &= P\left(Z \leq \frac{\frac{x - x_0 - \mu t}{\sigma} - 0}{\sqrt{t}}\right) = \Phi\left(\frac{x - x_0 - \mu t}{\sigma\sqrt{t}}\right). \end{aligned}$$

Differentiating this equation with respect to x yields

$$\begin{aligned} f_{t|0}(x|x_0) &= \frac{1}{\sigma\sqrt{t}} \cdot \frac{1}{\sqrt{2\pi}} \exp\left[-\left(\frac{x - x_0 - \mu t}{\sigma\sqrt{t}}\right)^2 / 2\right] \\ &= \frac{1}{\sigma\sqrt{2\pi t}} \exp\left[\frac{-(x - x_0 - \mu t)^2}{2\sigma^2 t}\right], \end{aligned}$$

which is the probability density function of a normal random variable with mean $x_0 + \mu t$ and variance $\sigma^2 t$.

Geometric Brownian Motion

As mentioned before, particles suspended in a liquid are constantly and incessantly under random collisions with nearby particles. As a result, they undergo unceasing motions in a totally erratic way, and the changes in each coordinate of the positions of a particle, in infinitesimal time periods, form a Brownian motion. In finance, random fluctuations of prices of shares of stocks, or other assets, are a result of an enormity of political, social, economical, and financial factors independently assessed by a large number of traders. For that reason financial scholars and consultants use Brownian motions as models for prices of assets in general and shares of stocks in particular. However, since such prices are non-negative and in the long run follow exponential growth or decay curves, they use a type of a stochastic process that is called **geometric Brownian motion**.

Let $\{W(t): t \geq 0\}$ be a standard Brownian motion. Let $X(t) = \mu t + \sigma W(t)$ so that $\{X(t): t \geq 0\}$ is a Brownian motion with drift parameter μ and variance parameter σ^2 . The process $\{V(t): t \geq 0\}$ defined by

$$V(t) = v_0 \exp[X(t)] = v_0 \exp[\mu t + \sigma W(t)]$$

is said to be a **geometric Brownian motion** with drift parameter μ and variance parameter σ^2 . Clearly, $W(0) = 0$ implies that $V(0) = v_0$.

Stock prices are often assumed to be Markov processes, since, usually, the price of a stock in the future depends on its present value only. The past history of the value of a stock and how it evolved to form the current price does not play a major role on the price of the stock in the future. In the preceding geometric model, if $V(t)$ is the price of some stock at t , then, for $t_0 < t_1 < \dots < t_n$, the relations

$$\frac{V(t_i)}{V(t_{i-1})} = \exp[\mu(t_i - t_{i-1})] \cdot \exp[\sigma(W(t_i) - W(t_{i-1}))]$$

show that

$$\frac{V(t_1)}{V(t_0)}, \frac{V(t_2)}{V(t_1)}, \dots, \frac{V(t_n)}{V(t_{n-1})},$$

the ratio changes of the price of the stock over nonoverlapping time intervals, are independent random variables.

We will now calculate the expected value and variance of $V(t)$ given that $V(0) = v_0$. To do so, let $Z \sim N(0, 1)$. We will first show that, for $\alpha \in (-\infty, \infty)$,

$$E(e^{\alpha Z}) = e^{\alpha^2/2}. \quad (21)$$

This follows from the following calculations:

$$\begin{aligned} E(e^{\alpha Z}) &= \int_{-\infty}^{\infty} e^{\alpha x} \cdot \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \\ &= e^{\alpha^2/2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\alpha^2 + \alpha x - \frac{1}{2}x^2} dx \\ &= e^{\alpha^2/2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-\alpha)^2} dx = e^{\alpha^2/2}, \end{aligned}$$

where $\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-\alpha)^2} dx = 1$, since $\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-\alpha)^2}$ is the probability density function of a normal random variable with mean α and variance 1.

Using (21) and the fact that $Z = W(t)/\sqrt{t} \sim N(0, 1)$, calculation of $E[V(t)]$ and $\text{Var}[V(t)]$ is straightforward. We have

$$E[V(t)] = E[v_0 e^{\mu t + \sigma W(t)}] = v_0 e^{\mu t} E(e^{\sigma \sqrt{t} Z}) = v_0 e^{\mu t + \frac{1}{2} \sigma^2 t}.$$

Also,

$$E[V(t)^2] = E[v_0^2 e^{2\mu t + 2\sigma W(t)}] = v_0^2 e^{2\mu t} E(e^{2\sigma \sqrt{t} Z}) = v_0^2 e^{2\mu t + 2\sigma^2 t}.$$

Therefore,

$$\text{Var}[V(t)] = E[V(t)^2] - E[V(t)]^2 = v_0^2 e^{2\mu t + \sigma^2 t} (e^{\sigma^2 t} - 1).$$

Example 12j Let $V(t)$ be the price of a stock, per share, at time t . Suppose that the stock's current value, per share, is \$40 with drift parameter \$2 per year and variance parameter 4.41. If $\{V(t) : t \geq 0\}$ is a geometric Brownian motion, what is the probability that after 18 months the stock price, per share, exceeds \$65?

Solution: Noting that $\sqrt{4.41} = 2.1$, we have

$$V(t) = 40 e^{2t + 2.1 W(t)},$$

where $\{W(t) : t \geq 0\}$ is a standard Brownian motion. Hence $W(t) \sim N(0, t)$. The desired probability is

$$\begin{aligned}
 P(V(1.5) > 65) &= P(40e^{2(1.5)+2.1W(1.5)} > 65) \\
 &= P(e^{2.1W(1.5)} > 0.0809) = P(W(1.5) > -1.197) \\
 &= P\left(\frac{W(1.5) - 0}{\sqrt{1.5}} > -\frac{1.197}{\sqrt{1.5}}\right) = P(Z > -0.98) \\
 &= 1 - \Phi(-0.98) = \Phi(0.98) = 0.8365. \quad \blacklozenge
 \end{aligned}$$

EXERCISES

- Suppose that liquid in a container is placed in a coordinate system, and at time 0, a pollen particle suspended in the liquid is at $(0, 0, 0)$, the origin. Let $Z(t)$ be the z -coordinate of the position of the pollen after t minutes. Suppose that $\{Z(t) : t \geq 0\}$ is a Brownian motion with variance parameter 9. Suppose that after 5 minutes the z -coordinate of the pollen's position is 0 again.
 - What is the probability that after 10 minutes it is between $-1/2$ and $1/2$?
 - If after seven minutes the z -coordinate of the pollen's position is -1 , find the expected value and variance of the z -coordinate of the position of pollen after six minutes.
- Let $\{X(t) : t \geq 0\}$ be a Brownian motion with variance parameter σ^2 . Show that, for all $t > 0$, $|X(t)|$ and $\max_{0 \leq s \leq t} X(s)$ are identically distributed.
- Let $\{X(t) : t \geq 0\}$ be a Brownian motion with variance parameter σ^2 . For $\varepsilon > 0$, show that $\lim_{t \rightarrow 0} P\left(\frac{|X(t)|}{t} > \varepsilon\right) = 1$, whereas $\lim_{t \rightarrow \infty} P\left(\frac{|X(t)|}{t} > \varepsilon\right) = 0$.
- Let $\{X(t) : t \geq 0\}$ be a Brownian motion with variance parameter σ^2 . Let T_α be the time of hitting α first. Let $Y \sim N(0, \sigma^2/\alpha^2)$. Show that, for $\alpha > 0$, T_α and $1/Y^2$ are identically distributed.
- Let $\{X(t) : t \geq 0\}$ be a Brownian motion with variance parameter σ^2 . For a fixed $t > 0$, let T be the smallest zero greater than t . Find the distribution function of T .
- Let $\{X(t) : t \geq 0\}$ be a Brownian motion with variance parameter σ^2 . As we know, for t_1 and t_2 , $t_1 < t_2$, the random variables $X(t_1)$ and $X(t_2)$ are not independent. Find the distribution of $X(t_1) + X(t_2)$.
- Let $\{X(t) : t \geq 0\}$ be a Brownian motion with variance parameter σ^2 . For $u > 0$, show that

$$E[X(t+u) \mid X(t)] = X(t).$$

Therefore, for $s > t$, $E[X(s) \mid X(t)] = X(t)$.

8. Let $\{X(t) : t \geq 0\}$ be a Brownian motion with variance parameter σ^2 . For $u > 0$, $t \geq 0$, find $E[X(t)X(t+u)]$.
9. **(Reflected Brownian Motion)** Suppose that liquid in a cubic container is placed in a coordinate system in such a way that the bottom of the container is placed on the xy -plane. Therefore, whenever a particle reaches the xy -plane, it cannot cross the bottom of the container. So it reverberates back to the nonnegative side of the z -axis. Suppose that at time 0, a particle is at $(0, 0, 0)$, the origin. Let $V(t)$ be the z -coordinate of the particle after t units of time. Find $E[V(t)]$, $\text{Var}[V(t)]$, and $P(V(t) \leq z \mid V(0) = z_0)$.

Hint: Let $\{Z(t) : t \geq 0\}$ be a Brownian motion with variance parameter σ^2 . Note that

$$V(t) = \begin{cases} Z(t) & \text{if } Z(t) \geq 0 \\ -Z(t) & \text{if } Z(t) < 0. \end{cases}$$

The process $\{V(t) : t \geq 0\}$ is called **reflected Brownian motion**.

10. Suppose that liquid in a cubic container is placed in a coordinate system. Suppose that at time 0, a particle is at $(0, 0, 0)$, the origin. Let $(X(t), Y(t), Z(t))$ be the coordinates of the particle after t units of time, and assume that $X(t)$, $Y(t)$, and $Z(t)$ are independent Brownian motions, each with variance parameter σ^2 . Let $D(t)$ be the distance of the particle from the origin after t units of time. Find $E[D(t)]$.
11. Let $V(t)$ be the price of a stock, per share, at time t . Suppose that the stock's current value, per share, is \$95.00 with drift parameter $-\$2$ per year and variance parameter 5.29. If $\{V(t) : t \geq 0\}$ is a geometric Brownian motion, what is the probability that after 9 months the stock price, per share, is below \$80?
12. Let $\{X(t) : t \geq 0\}$ be a Brownian motion with variance parameter σ^2 . For $t, s > 0$, find $E[X(s)X(t)]$. Use the fact that, by Exercise 7, for $s < t$,

$$E[X(t) \mid X(s)] = X(s).$$



† Self-Quiz on Section 12B

Time allotted: 15 Minutes

1. Suppose that liquid in a cubic container is placed in a coordinate system, and at time 0, a pollen particle suspended in the liquid is at $(0, 0, 0)$, the origin. Let $Y(t)$ be the y -coordinate of the position of the pollen after t minutes. Suppose that the process $\{Y(t) : t \geq 0\}$ is a Brownian motion with variance parameter σ^2 . For $0 < s < t$, find $E[Y(s) \mid Y(t) = y]$ and $\text{Var}[Y(s) \mid Y(t) = y]$. (3 points)
2. Let $V(t)$ be the price of a stock, per share, at time t . Suppose that $\{V(t) : t \geq 0\}$ is a geometric Brownian motion with drift parameter \$3 per year and variance parameter 27.04. What is the probability that the price of this stock is at least twice its current price after two years? (4 Points)
3. Let $\{X(t) : t \geq 0\}$ be a Brownian motion with variance parameter σ^2 . For $t > 0$, let T be the smallest zero greater than t , and let U be the largest zero smaller than t . For $x < t < y$, find $P(U < x \text{ and } T > y)$. (3 points)



SECTION 12B SUMMARY

► A stochastic process $\{X(t) : t \geq 0\}$ with state space $\mathcal{S} = (-\infty, \infty)$ that possesses stationary and independent increments is said to be a *Brownian motion* with *variance parameter* σ^2 if $X(0) = 0$ and, for $t > 0$, $X(t)$ is a normal random variable with mean 0 and variance $\sigma^2 t$, for some $\sigma > 0$. For such a process, by stationarity of the increments, for $s, t \geq 0$, $X(t+s) - X(s)$ is $N(0, \sigma^2 t)$. A Brownian motion with variance parameter 1 is called a *standard Brownian motion*.

► For a Brownian motion $\{X(t) : t \geq 0\}$, $f_{t|0}(x|x_0)$, the conditional probability density function of $X(t)$ given that $X(0) = x_0$ is normal with mean x_0 and standard deviation $\sigma\sqrt{t}$.

► For a Brownian motion $\{X(t) : t \geq 0\}$, for $t_1 < t_2$, $f(x_1, x_2)$ the joint probability density function of $X(t_1)$ and $X(t_2)$ given by

$$f(x_1, x_2) = \frac{1}{2\sigma^2\pi\sqrt{t_1(t_2 - t_1)}} \exp\left[-\frac{1}{2\sigma^2}\left(\frac{x_1^2}{t_1} + \frac{(x_2 - x_1)^2}{t_2 - t_1}\right)\right], \quad -\infty < x_1, x_2 < \infty.$$

†For the solutions to these Self-Quiz problems, see Solutions to Self-Quizzes and Self-Tests, Chapter 12, of this Companion Website.

For $t_1 < t_2 < \cdots < t_n$, $f(x_1, x_2, \dots, x_n)$, the joint probability density function of $X(t_1)$, $X(t_2), \dots, X(t_n)$, is

$$f(x_1, x_2, \dots, x_n) = \frac{\exp \left[-\frac{1}{2\sigma^2} \left(\frac{x_1^2}{t_1} + \sum_{i=1}^{n-1} \frac{(x_{i+1} - x_i)^2}{t_{i+1} - t_i} \right) \right]}{\sigma^n \sqrt{(2\pi)^n t_1(t_2 - t_1) \cdots (t_n - t_{n-1})}}, \quad -\infty < x_1, x_2, \dots, x_n < \infty.$$

► For a Brownian motion $\{X(t) : t \geq 0\}$,

$$E[X(u) \mid X(0) = 0 \text{ and } X(u_1) = 0] = 0,$$

$$\text{Var}[X(u) \mid X(0) = 0 \text{ and } X(u_1) = 0] = \frac{\sigma^2 u(u_1 - u)}{u_1}.$$

Furthermore, for $t_1 < t < t_2$, the conditional probability density function of $X(t)$ given that $X(t_1) = x_1$ and $X(t_2) = x_2$ is normal with mean $x_1 + \frac{x_2 - x_1}{t_2 - t_1}(t - t_1)$ and variance $\sigma^2 \frac{(t_2 - t)(t - t_1)}{t_2 - t_1}$.

► Let $\{X(t) : t \geq 0\}$ be a Brownian motion with variance parameter σ^2 . The *first passage time* to a point α , also called the *time of hitting α first*, denoted by T_α , is the minimum of the set $\{t : X(t) = \alpha\}$. We have that for all $\alpha \in (-\infty, \infty)$, the distribution function of the first passage time to α is given by

$$P(T_\alpha \leq t) = 2 \left[1 - \Phi \left(\frac{|\alpha|}{\sigma \sqrt{t}} \right) \right].$$

Note that, even though, with probability 1, $\{X(t) : t \geq 0\}$ eventually will hit α , none of the moments of the first passage time to α is finite.

► As before, let T_α be the first passage time to α . By the continuity of the paths of Brownian motions, for $\alpha > 0$, the event $\max_{0 \leq s \leq t} X(s) \geq \alpha$ occurs if and only if $T_\alpha \leq t$. This observation enables us to calculate the distribution function of $\max_{0 \leq s \leq t} X(s)$:

$$P\left(\max_{0 \leq s \leq t} X(s) \leq x\right) = \begin{cases} 2\Phi\left(\frac{x}{\sigma \sqrt{t}}\right) - 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0. \end{cases}$$

► **The Zeros of Brownian Motion** For a Brownian motion $\{X(t) : t \geq 0\}$ with variance parameter σ^2 , the probability that $X(t) = 0$ at least once in the time interval (t_1, t_2) , $0 < t_1 < t_2$, is $\frac{2}{\pi} \arccos \sqrt{\frac{t_1}{t_2}}$.

► **Brownian Motion with Drift** Let $\{W(t): t \geq 0\}$ be a standard Brownian motion. That is, a Brownian motion with variance parameter 1. For $\mu \in (-\infty, \infty)$ and $\sigma > 0$, the process

$$X(t) = \mu t + \sigma W(t), \quad t \geq 0,$$

is called the *Brownian motion with drift parameter μ and variance parameter σ^2* . Note that for such a process, it is still the case that $X(0) = 0$ and $\{X(t): t \geq 0\}$ possesses stationary and independent increments. However, for $t > 0$, $X(t)$ is normal with mean μt and variance $\sigma^2 t$. For a Brownian motion $\{X(t): t \geq 0\}$ with drift parameter μ and variance parameter σ^2 , let $f_{t|0}(x|x_0)$ be the conditional probability density function of $X(t)$ given that $X(0) = x_0$, or, equivalently, the conditional probability density function of $X(t + t_0)$ given that $X(t_0) = x_0$. Then $f_{t|0}(x|x_0)$ is the probability density function of a normal random variable with mean $x_0 + \mu t$ and variance $\sigma^2 t$.

► **Geometric Brownian Motion** Let $\{W(t): t \geq 0\}$ be a standard Brownian motion. Let $X(t) = \mu t + \sigma W(t)$ so that $\{X(t): t \geq 0\}$ is a Brownian motion with drift parameter μ and variance parameter σ^2 . The process $\{V(t): t \geq 0\}$ defined by

$$V(t) = v_0 \exp [X(t)] = v_0 \exp [\mu t + \sigma W(t)]$$

is said to be a *geometric Brownian motion* with drift parameter μ and variance parameter σ^2 . We have

$$E[V(t)] = v_0 e^{\mu t + \frac{1}{2}\sigma^2 t} \quad \text{and} \quad \text{Var}[V(t)] = v_0^2 e^{2\mu t + \sigma^2 t} (e^{\sigma^2 t} - 1).$$

Answers to Odd-Numbered Exercises of Section 12B

- 1.** (a) 0.056; (b) -0.5 and 4.5 , respectively.
- 5.** 0, if $x \leq t$; $(2/\pi) \arccos \sqrt{t/x}$, if $x \geq t$.
- 9.** $\sigma\sqrt{2t/\pi}$, $\sigma^2 t[1 - (2/\pi)]$, and $\Phi[(z + z_0)/\sigma\sqrt{t}] + \Phi[(z - z_0)/\sigma\sqrt{t}] - 1$, respectively.
- 11.** 0.7486.